



PHEMTO: MEGAlib Simulations ^{v2}

José Sousa - CEA/Dap

22-01-2026

https://github.com/jcfsousa/phemto_simulations

Overall Assumptions

Detectors

https://github.com/jcfsousa/phemto_simulations

CdTe Detector performance:

density = 5.79 g/cm³

thickness = 2mm

threshold = 4 keV

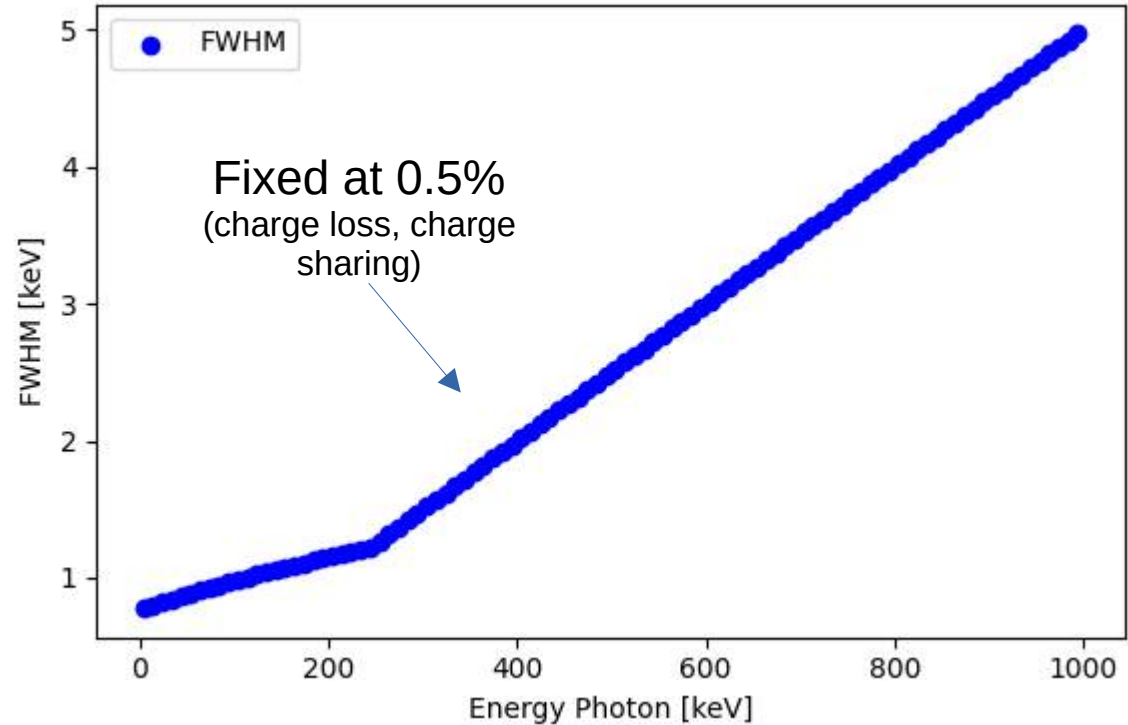
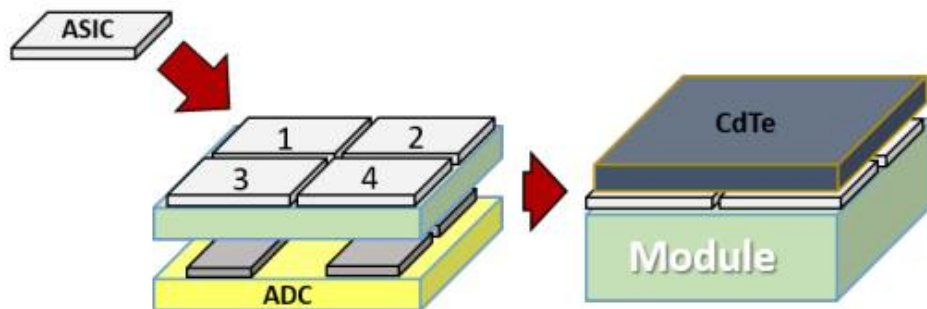
dx dy = 0.25mm

Energy resolution: 74e rms

PHEMTO config:

4x4 modules (6.4 x 6.4 cm²)

256 x 256 pix



1 module: 1.4cm, 64x64 pix

Overall Assumptions

Detectors

https://github.com/jcfsousa/phemto_simulations

Si Detector performance:

density = 5.79 g/cm³

thickness = 450μm

threshold = 2 keV

dx dy = 0.13mm

Energy resolution: 0.35keV FWHM

PHEMTO config:

6.65 x 6.65 cm²

512 x 512 pix

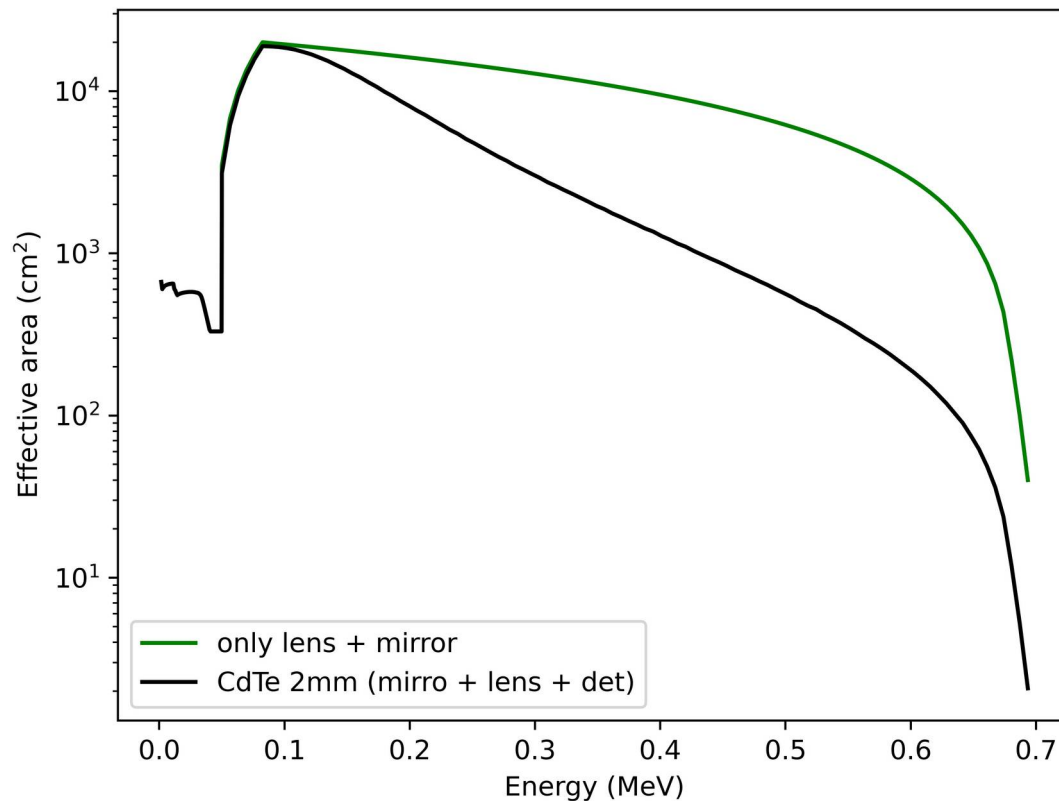
```
// Si strip detector information:
Voxel3D                SiSDDet
SiSDDet.SensitiveVolume Si
SiSDDet.VoxelNumber     512 512 1
SiSDDet.Offset          0 0 0

// Threshold in keV
SiSDDet.NoiseThreshold  0.01
SiSDDet.TriggerThreshold 2.
SiSDDet.EnergyResolution Gauss 4.0 4.0 0.15
SiSDDet.EnergyResolution Gauss 6.0 6.0 0.15
```

Overall Assumptions

Mirror Lenses

https://github.com/jcfsousa/phemto_simulations



Overall Assumptions

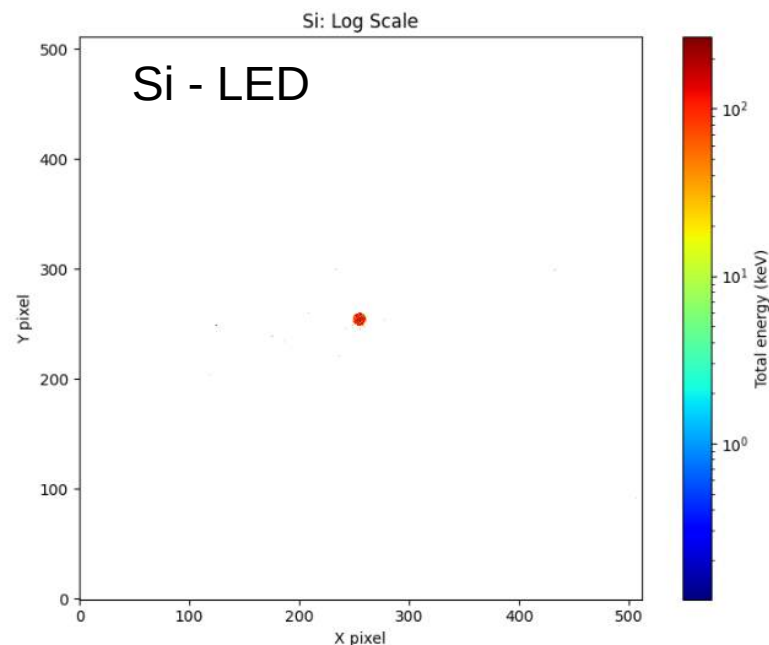
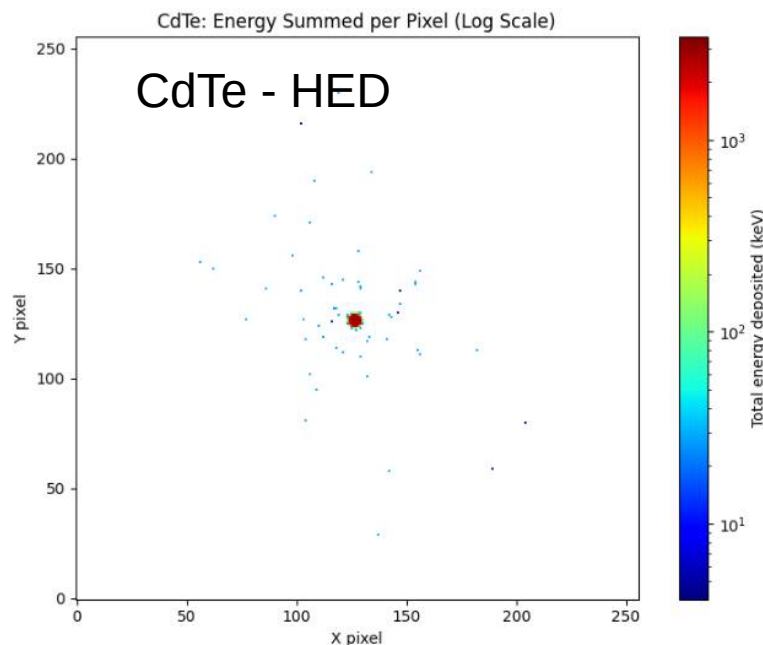
PSF

https://github.com/jcfsousa/phemto_simulations

Laue Lense approximation:

- Homogeneous
- 1.5mm diameter focus [1,2]

*Used this approximation also for
x-ray mirror PSF, [1, 400]keV*



Overall Assumptions

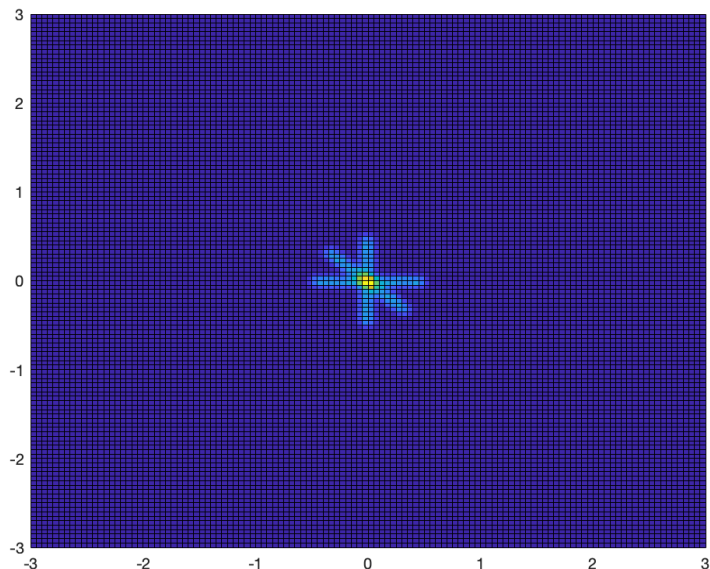
PSF

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Laue Lense approximation:

- Homogeneous
- 1.5mm diameter focus [1,2]

Laue Lenses Future MEGAlib source:



*Used this approximation also for
x-ray mirror PSF, [1, 400]keV*

X-ray mirror Future MEGAlib source:



Miguel Moite + Lisa Ferro

Instrument Configuration

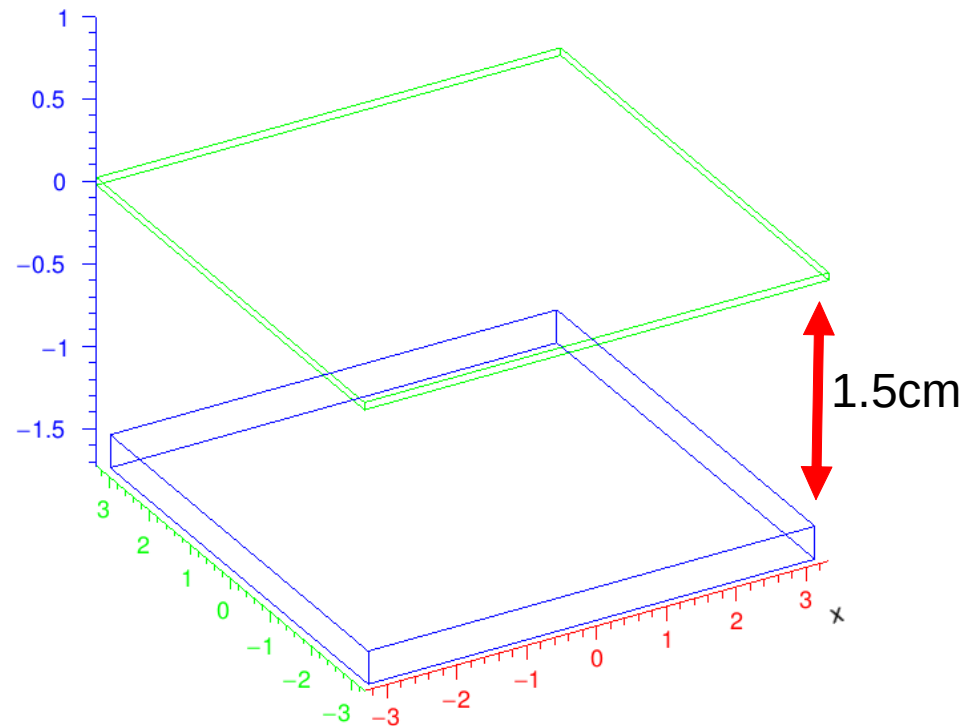
MEGAlib Model

https://github.com/jcfsousa/phemto_simulations

Si (green):
6.65x6.65cm², 450um thick

CdTe (blue):
6.4x6.4cm², 2mm thick

1.5cm spacing - Instrument
team worried about
mechanical spacing

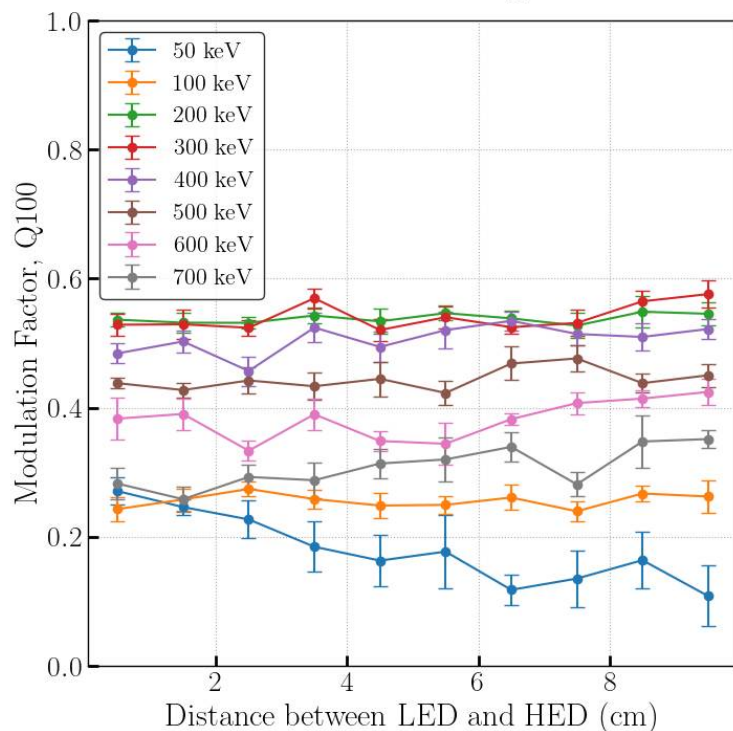


Instrument Configuration

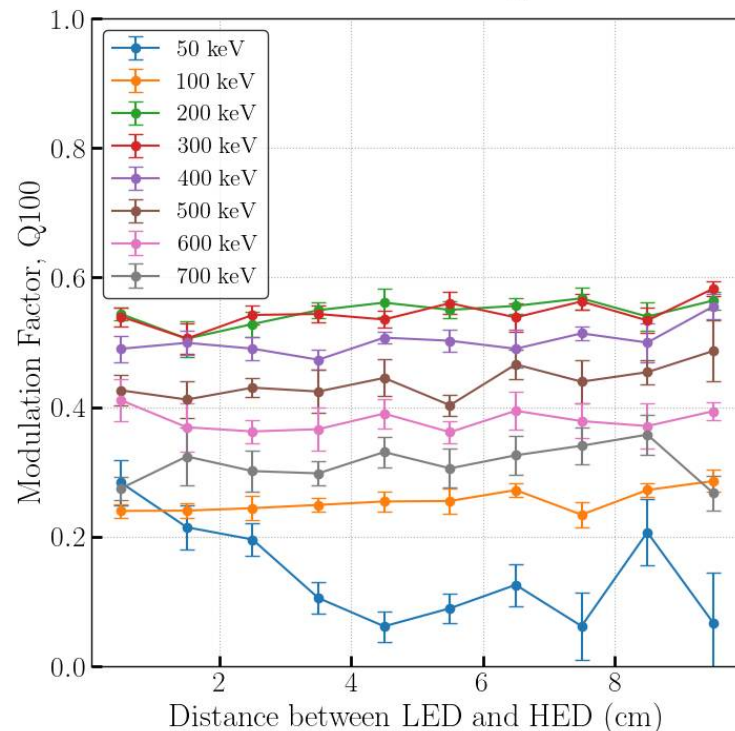
Larger HED?

https://github.com/jcfsousa/phemto_simulations

HED detector config 7x7



HED detector config 4x4

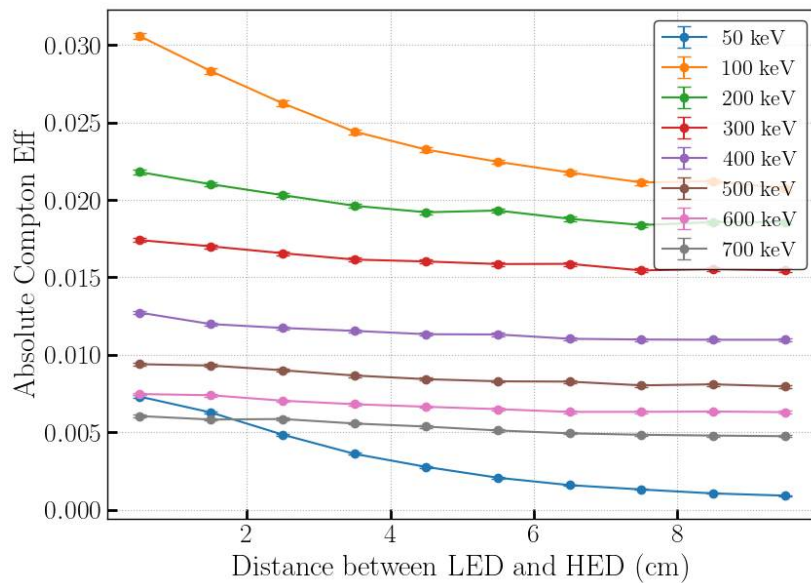


Instrument Configuration

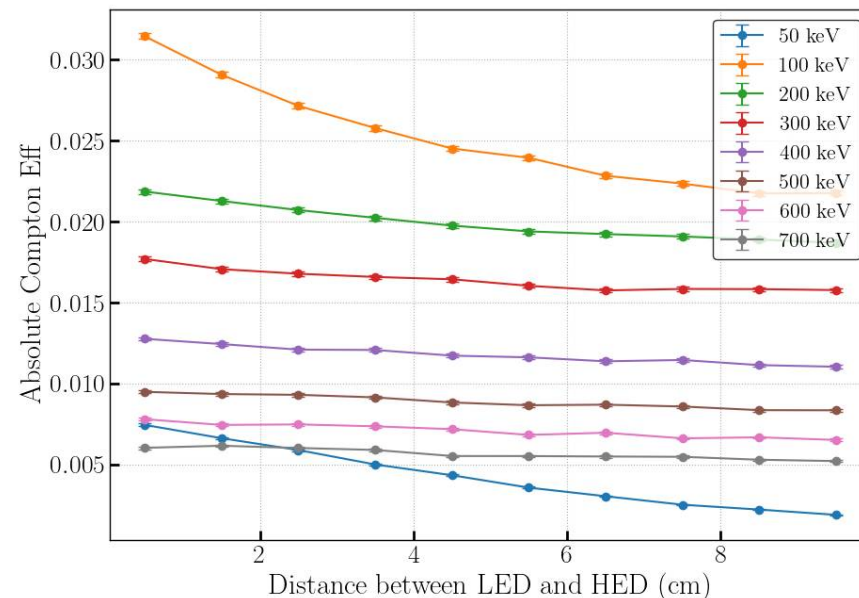
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HED detector config 4x4



HED detector config 7x7

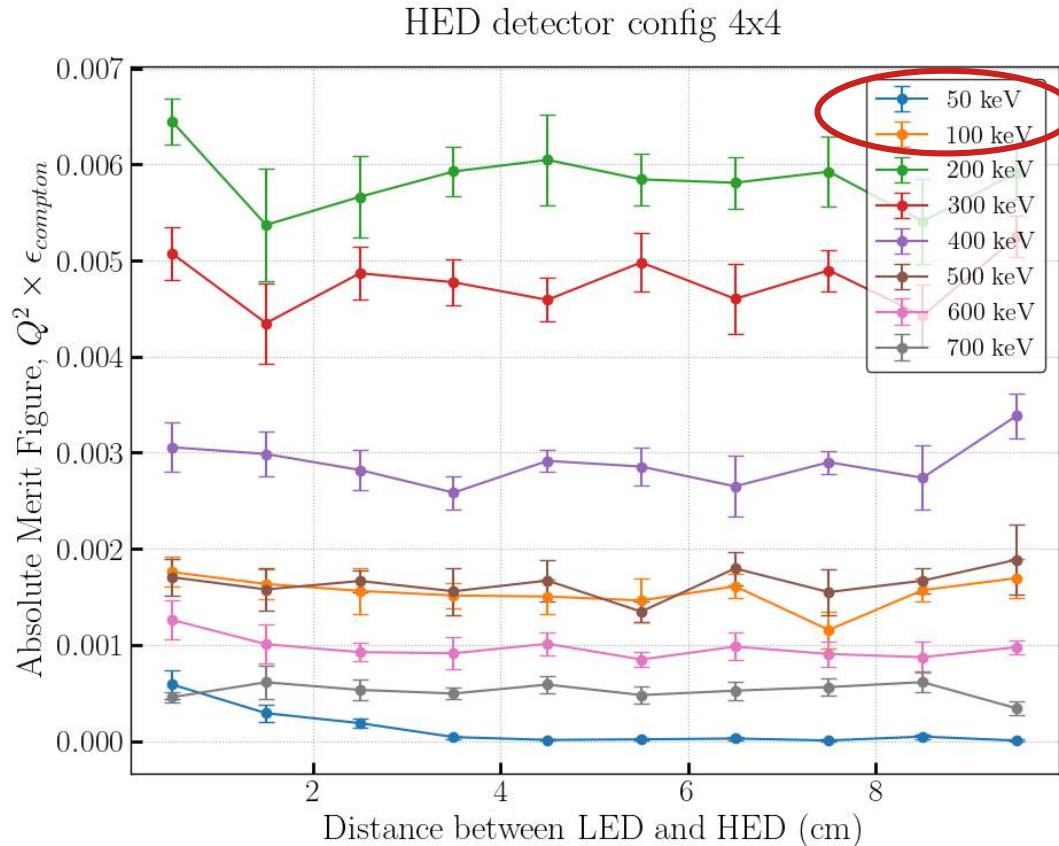


No Significant Impact on Compton Eff or Q with larger HED, 7x7

Instrument Configuration

Distance between LED - HED?

https://github.com/jcfsousa/phemto_simulations



$$MDP = \frac{a_S}{M} = \frac{4.29}{M R_S} \left[\frac{R_S + R_B}{T} \right]^{1/2}$$

$\uparrow$ $\uparrow$
Q Q

$$R_S = \text{Compt}_{\text{eff}} \cdot N_S$$

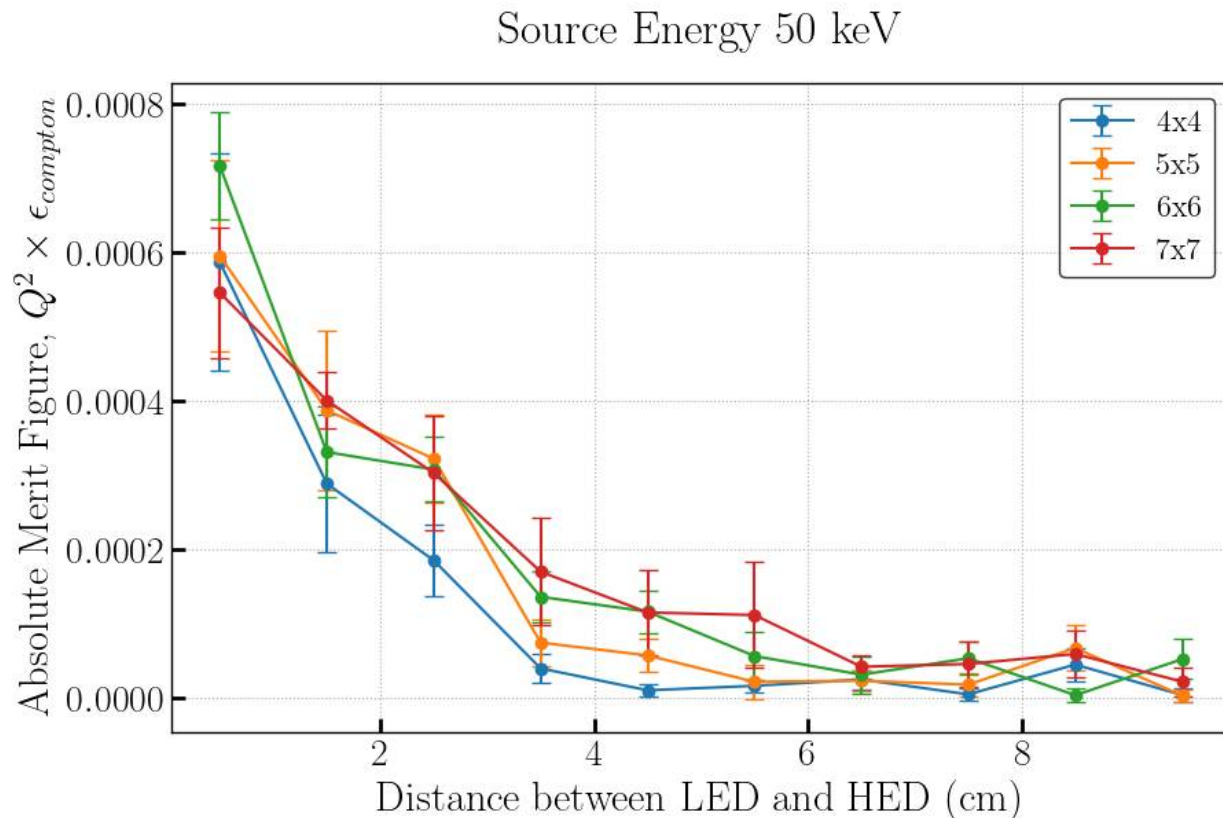
$$R_B = \text{Compt}_{\text{eff}} \cdot N_B$$

$$\text{Merit Figure} = Q^2 \cdot \text{Compt}_{\text{eff}}$$

Instrument Configuration

Distance between LED - HED?

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$$MDP = \frac{a_S}{M} = \frac{4.29}{M R_S} \left[\frac{R_S + R_B}{T} \right]^{1/2}$$

$\uparrow$ $\uparrow$
Q Q

$$R_S = \text{Compt}_{\text{eff}} \cdot N_S$$

$$R_B = \text{Compt}_{\text{eff}} \cdot N_B$$

$$\text{Merit Figure} = Q^2 \cdot \text{Compt}_{\text{eff}}$$

Instrument Configuration

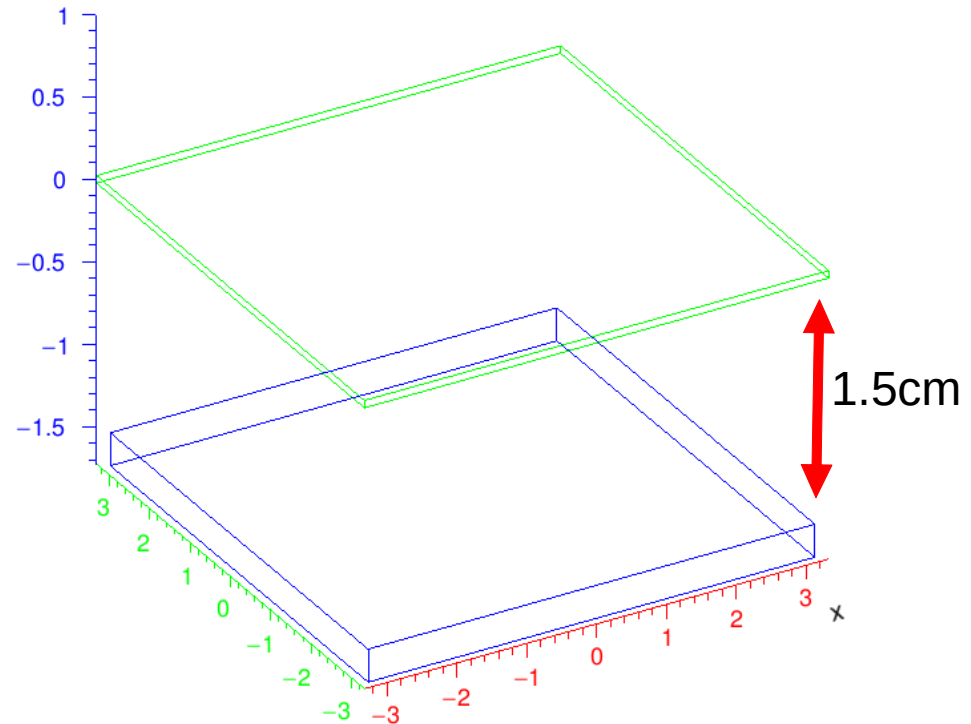
MEGAlib model

https://github.com/jcfsousa/phemto_simulations

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Efficiency and Sensitivity

PHEMTO

https://github.com/jcfsousa/phemto_simulations

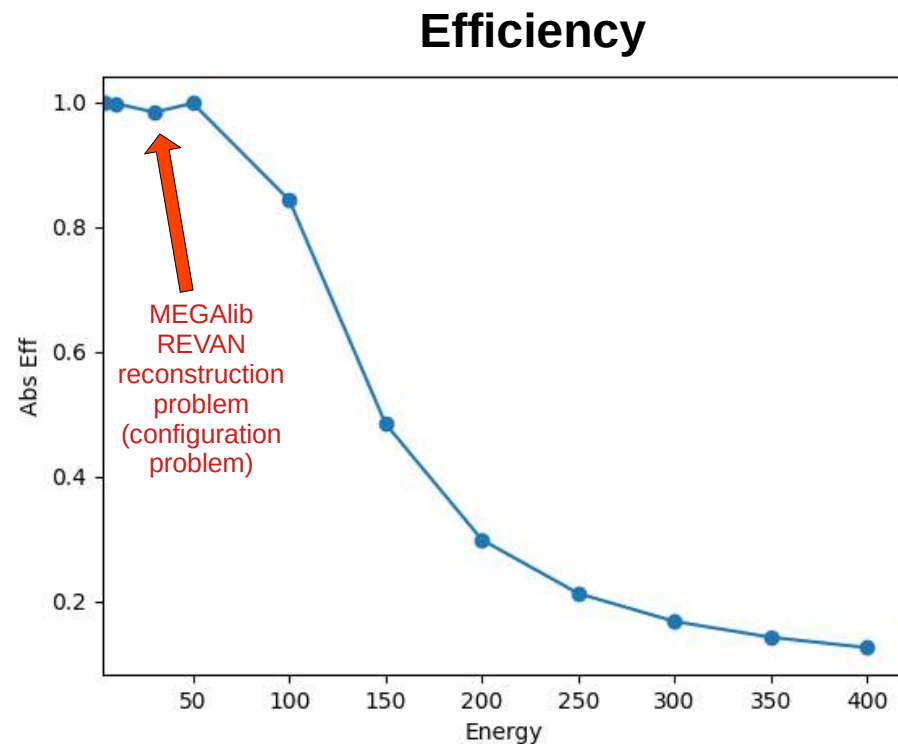
Simulated in MEGAlib instrument Efficiency

Didn't have time to simulate Background on MEGAlib.

Ignoring Particle background (Anti-coincidence)

Used Phillipe Laurent background approximation (seems to be close to reality for photons)

Assuming background only on PSF area



Efficiency and Sensitivity

PHEMTO

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Simulated in MEGAlib instrument Efficiency

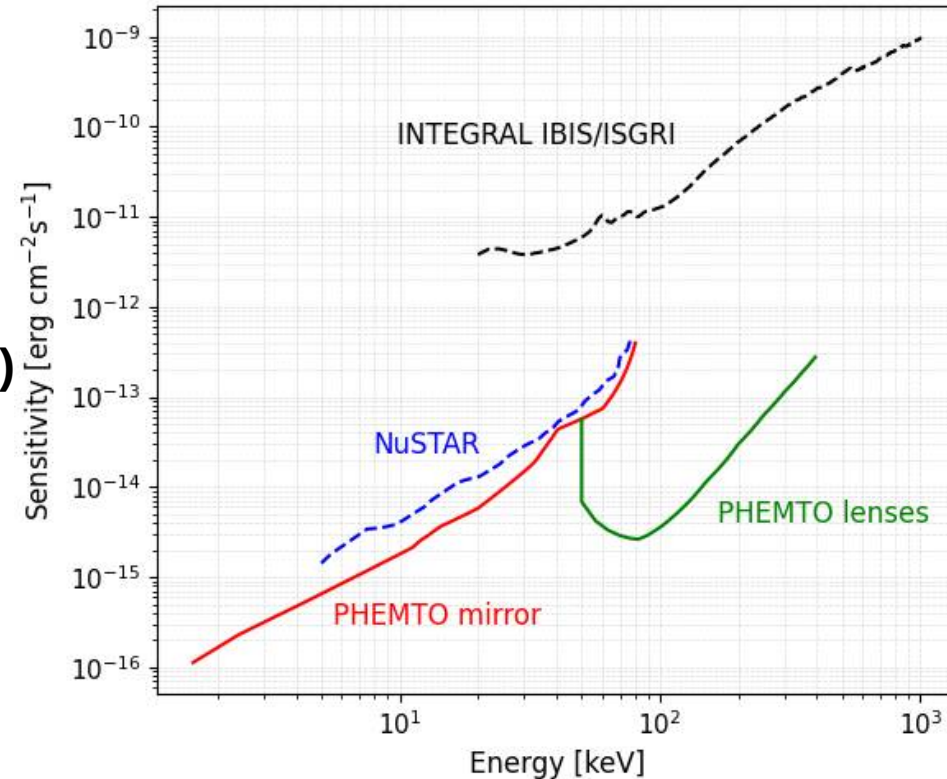
Didn't have time to simulate Background on MEGAlib.

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Assuming background only on PSF area

Sensitivity



Polarization

PHEMTO MDP

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Crab Flux [8]:

$$F(E) = K \left(\frac{E}{1 \text{ keV}} \right)^{-\alpha}$$

$$\alpha = 2.169 \pm 0.008$$

$$K = 14.44 \pm 0.44 \text{ ph/cm}^2/\text{s/keV}$$

Collimated beam, 1.5mm
diameter (PSF)

Azimuthal double
Compton distribution

Q100, Compt_eff

MEGALib Cosima
(Geant4)

100% polarized .source
Random polarized .source

Bin Energy
(ie: [50,400]keV or
[50,150]keV)

MEGALib

Revan

Event reconstruction
(Compton Chi2 via energies)

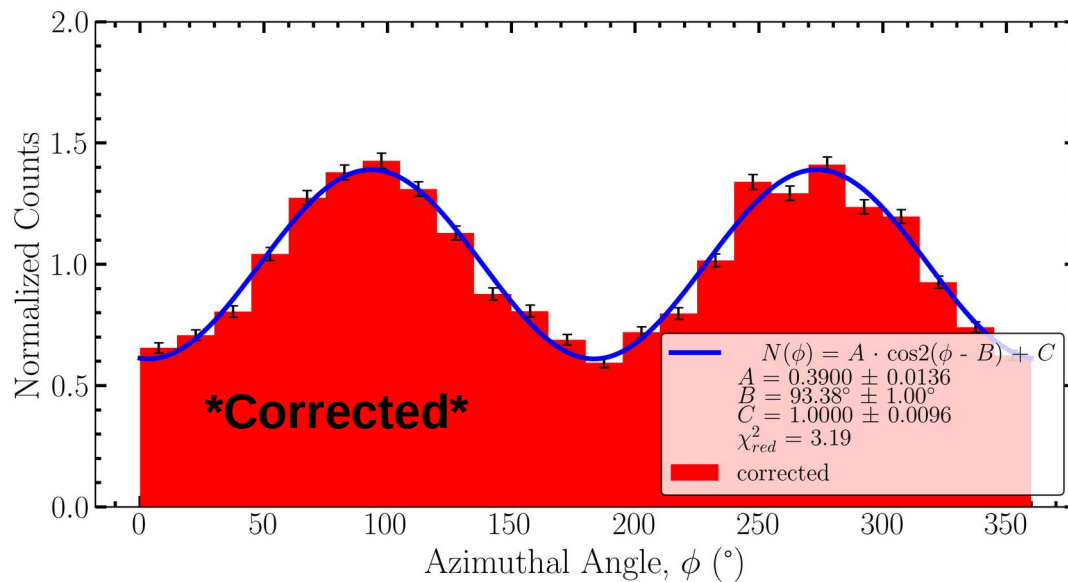
Parse .tra files + apply
Dets pixelization

Polarization

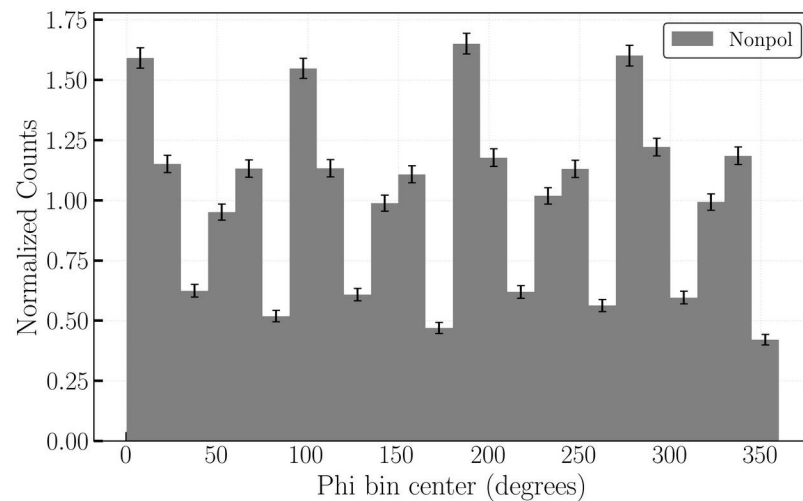
PHEMTO MDP

Crab Source, Ebin [50,400]keV

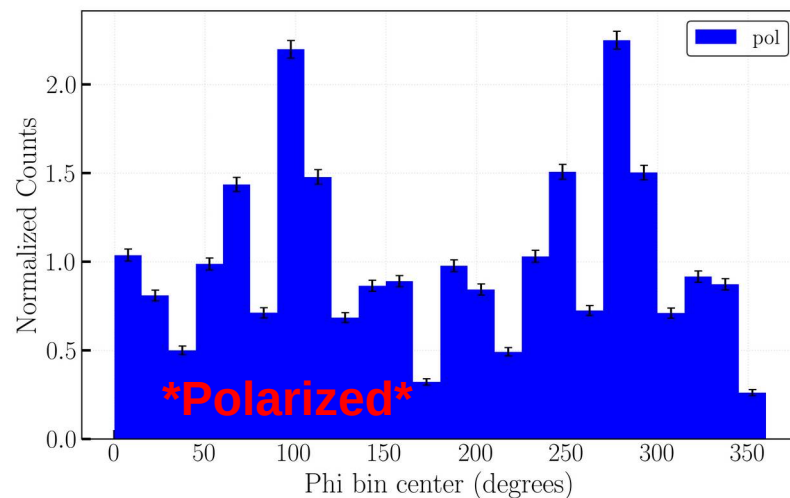
Crab1MeventsPol_50-400keV_config4x4.1.5cm keV



etry: NonPolarized Source (Energy: Crab1MeventsPol_50-400keV_config4x4.1.5cm keV)



etry: Polarized Source (Energy: Crab1MeventsPol_50-400keV_config4x4.1.5cm keV)

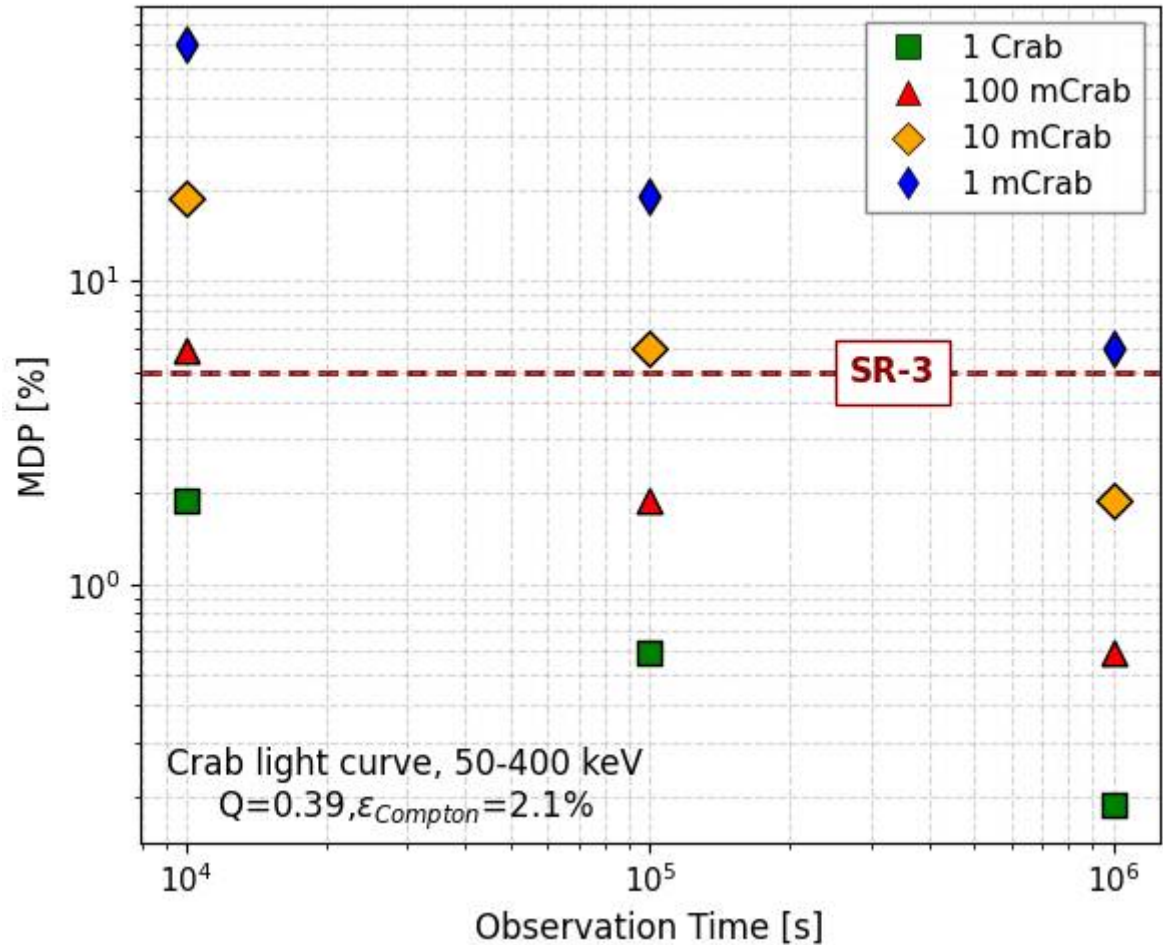


Polarization

PHEMTO MDP

https://github.com/jcfsousa/phemto_simulations

Crab Source
Ebin [50,400]keV

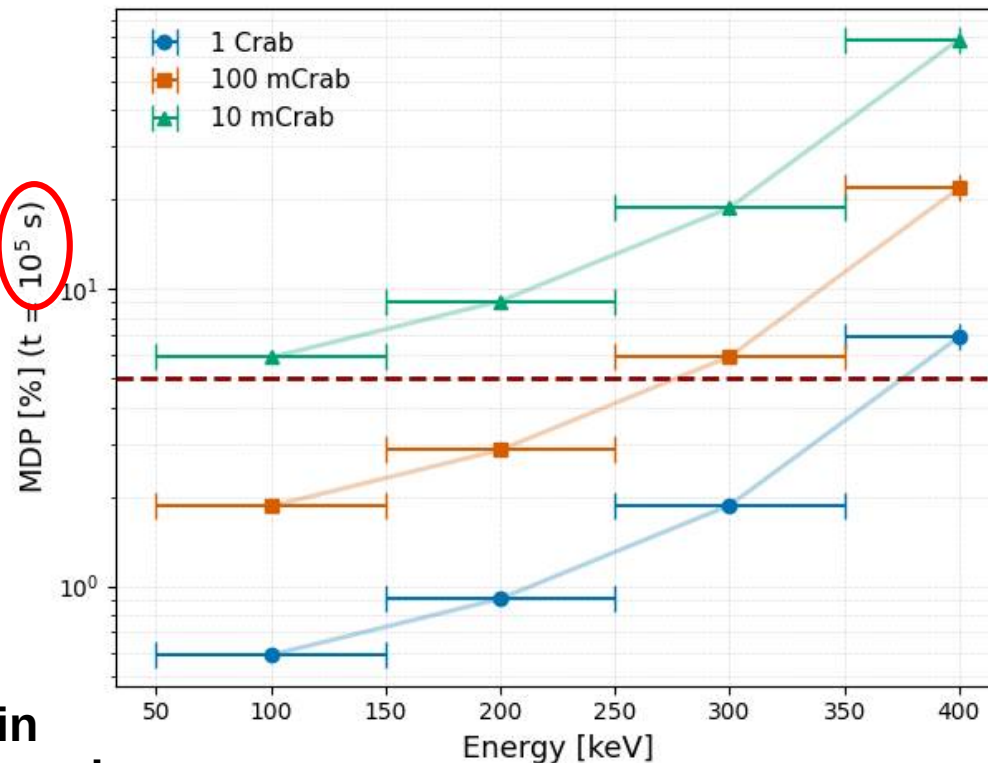


Polarization

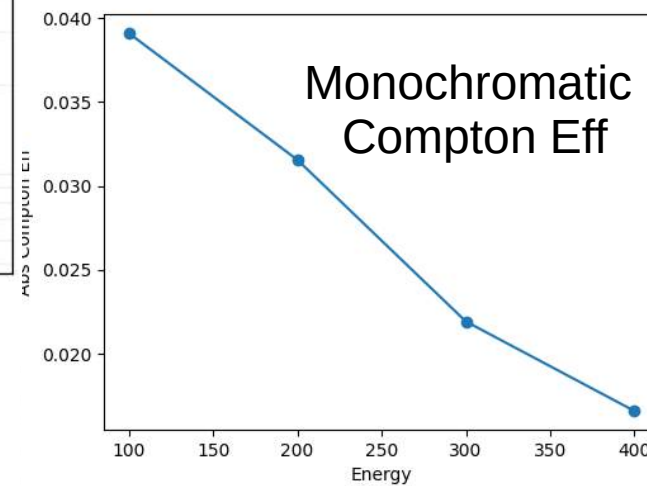
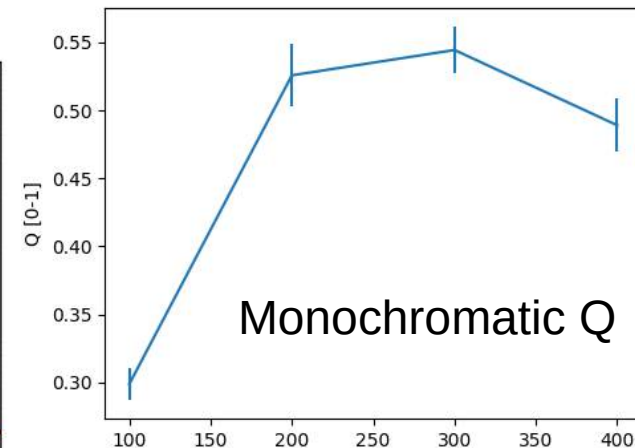
PHEMTO MDP

https://github.com/jcfsousa/phemto_simulations

Crab Source
Ebin 100 keV



*Used center bin
monochromatic Q and
Compton Eff*



Polarization

PHEMTO MDP - Background

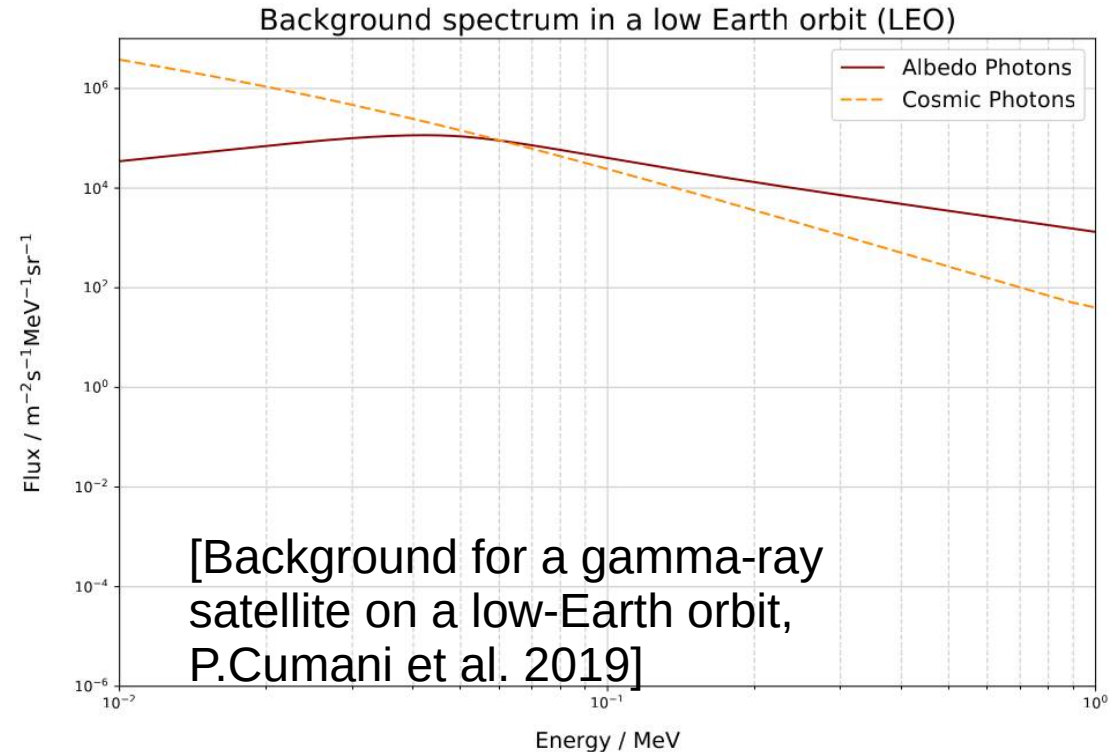
https://github.com/jcfsousa/phemto_simulations

Considered:

Cosmic Photons

Photoelectric on PSF

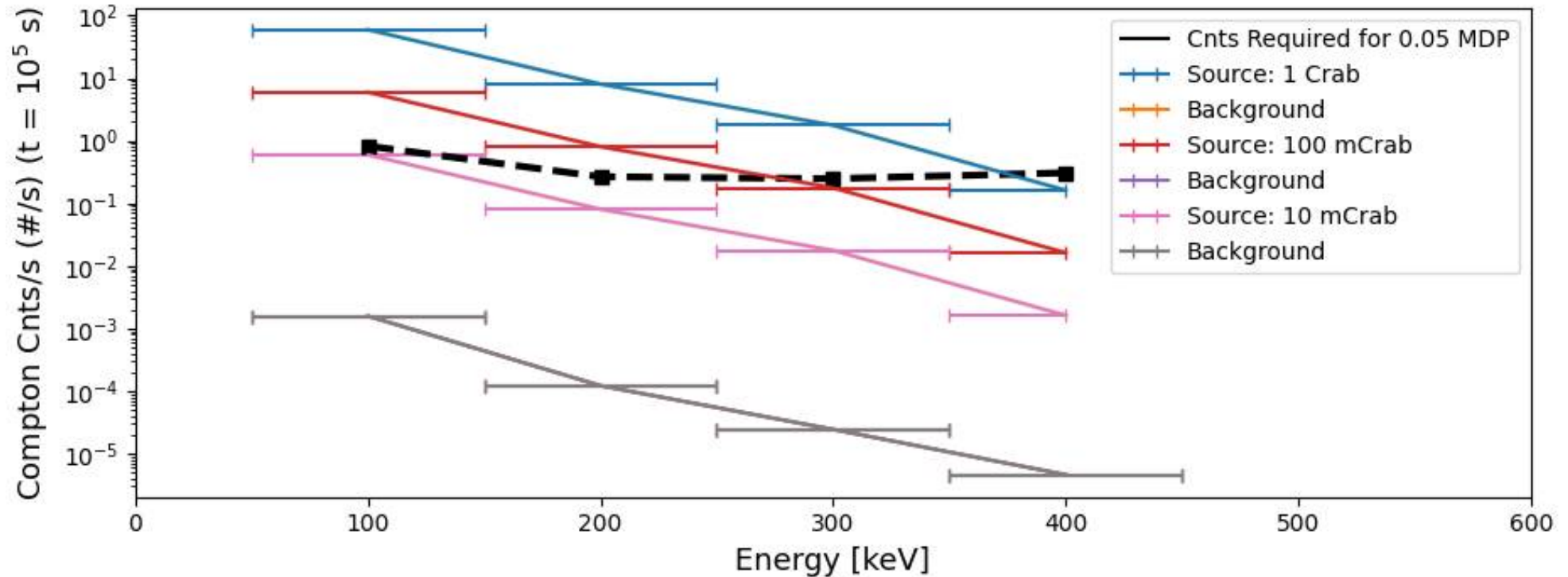
Compton with at least 1 event in PSF



Polarization

PHEMTO MDP - Background

https://github.com/jcfsousa/phemto_simulations



Future Work

https://github.com/jcfsousa/phemto_simulations

Detectors

- Better energy resolution information detectors
- Can we do Polarimetry with Si?

Improve MEGAlib geometry:

- Anti-Coincidence detector
- Shielding (collimator + space craft backside detectors
- Collimator

Background:

- Run Cosmic Background with collimator (urgent for Phase 0 sensitivity)
- Define orbit
- Add particle background, Cumani paper
- Activation

Optics:

- Laue lenses PSF and updated Effective Area
- X-ray mirror PSF and updated Effective Area



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References

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- [1] Ferrari, E., Ferrari, C., Virgilli, E., Caroli, E., Auricchio, N., Stephen, J. B., Ferro, L., da Silva, R. C., and Moita, M. (2025). Correction: Imaging performance of laue lenses made of ge and si bent crystals for future gamma-ray astrophysics telescopes. Aerotecnica Missili amp; Spazio.
- [2] Frontera, F. (2025). Hard x-ray/soft gamma-ray laue lenses for high energy astrophysics.
- [3] Lucchetta, G. et. al. (2022). Introducing the MeVCube concept: a CubeSat for MeV observations. [doi:10.1088/1475-7516/2022/08/013](https://doi.org/10.1088/1475-7516/2022/08/013)
- [4] <https://doi.org/10.48550/arXiv.2108.02860>
- [5] Cumani, P., Hernanz, M., Kiener, J. et al. Background for a gamma-ray satellite on a low-Earth orbit. Exp Astron 47, 273–302 (2019). <https://doi.org/10.1007/s10686-019-09624-0>

References

https://github.com/jcfsousa/phemto_simulation

[6] <https://doi.org/10.1051/0004-6361/202245205>

[7] **On understanding the figures of merit for detection and measurement of x-ray polarization**

Martin C. Weisskopf, Ronald F. Elsner, Stephen L. O'Dell <https://arxiv.org/abs/1006.3711>

[8] THE INTEGRAL/SPI RESPONSE AND THE CRAB OBSERVATIONS
arXiv:astro-ph/0406058v1 2 Jun 2004

[9] Andreas Zoglauer et al., "MEGAlib - Medium Energy Gamma-ray Astronomy Library", NewAR 50 (7-8), 2006